DELHI PUBLIC SCHOOL JANKIPURAM , LUCKNOW

SESSION 2025-2026: PRE BOARD - I

SUBJECT: SCIENCE(086)

SET–I

Time Allowed: 3 hours **M.M: 80**

Date : _____ **Name :** _____ **Class/SecX**____

General Instructions:

Read the following instructions carefully and follow them:

This question paper consists of 39 questions in 3 sections.

Section A is Biology, Section B is Chemistry and Section C is Physics.

All questions are compulsory. However, an internal choice is provided in some questions. A student is expected to attempt only one of these questions.

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Section – A

1. Secretion of less saliva in mouth will affect the conversion of : (1)

A. proteins into amino acids

B. fats into fatty acids and glycerol

C. starch into simple sugars

D. sugars into alcohol

2. The breakdown of glucose has taken the following pathway : Glucose (a) ⟶ Pyruvate + Energy (b) ⟶ Lactic acid + Energy The sites 'a' and 'b' respectively are : (1)

A. Mitochondria and Oxygen deficient muscle cells

B. Cytoplasm and Oxygen rich muscle cells

C. Cytoplasm and Yeast cells

D. Cytoplasm and Oxygen deficient muscle cells

3.A person accidentally touches a hot object and withdraws his hand instantly. Which of the following pathways is involved in this response? (1)

A. Brain → Spinal cord → Muscle

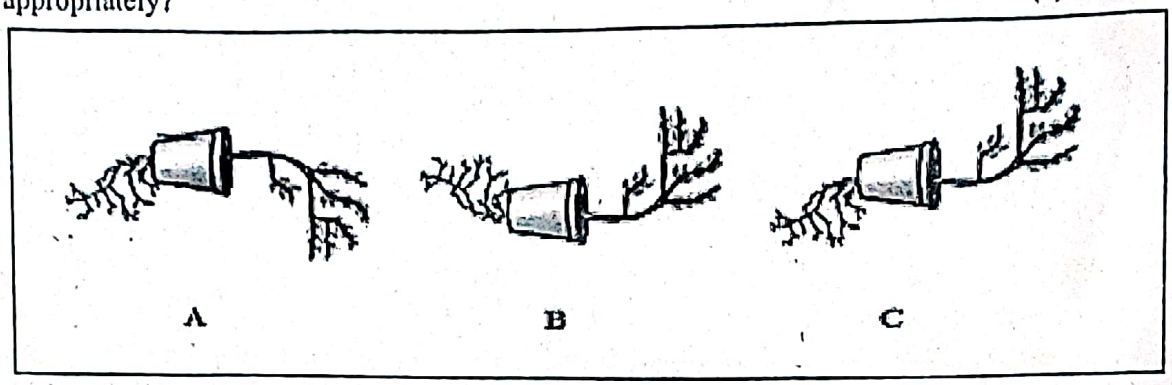
B. Spinal cord → Muscle → Brain

C. Receptor → Spinal cord → Effector

D. Receptor → Brain → Effector

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4. Observe the three figures given below. Which of the following depicts tropic movements appropriately? (1)



A. B and C

B. A and C

C. B only

D. C only

5. Which of the following techniques is used as a contraceptive method in males only? (1)

A. Copper-T

B. Oral pills

C. Blockage of fallopian tube

D. Blockage of vas deferens

6. Disposable plastic plates should not be used because: (1)

A. they are made of non-biodegradable materials

B. they are made of biodegradable materials

C. they are made of toxic materials

D. they are made of materials with light weight

7. Which of the following best explains why food webs are more stable than simple food chains?(1)

A. They have fewer organisms

B. They show only producers and consumers

C. They allow multiple feeding relationships

D. They depend on a single energy source

The following two questions consist of two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below:

(a) Both A and R are true, and R is the correct explanation of A.

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(b) Both A and R are true, and R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

8. Assertion (A): A geneticist crossed two pea plants and got 50% tall and 50% dwarf in the progeny. (1)

Reason (R): One plant was heterozygous tall and the other was dwarf.

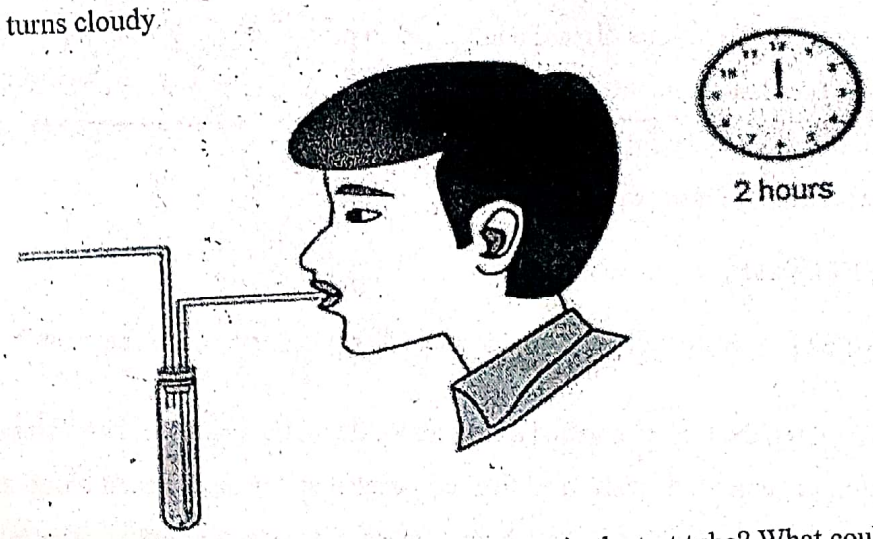
9. Assertion (A): Garden is an artificial ecosystem.

Reason (R): Biotic and abiotic components are manipulated by humans in the garden. (1)

10. Though liver does not secrete any digestive enzymes, even then it is an essential organ of digestive system. Explain. (2)

11. Attempt either option A or B

(A) Given below is an image of an experiment conducted by a student to understand the process of respiration. He blows into clear solution present in the test tube and sees that it turns cloudy. (2)



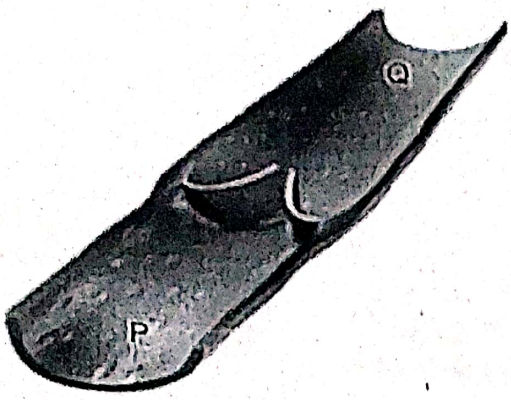
(i) What is the most likely substance present in the test tube? What could be the aim of this experiment?

(ii) What kind of respiration is shown in the experiment? Justify your answer.

OR

(B) The image below shows the cross-section of a blood vessel in a human arm. (2)

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(i) What is the type of blood vessel shown in the image? Write any two specific features of it.

(ii) Which direction will the blood flow in such a blood vessel?

12. (a) What will happen if we kill all organisms in one trophic level?

(b) Calculate the amount of energy available to the organisms at the fourth trophic level. If the energy available to the organisms at the second trophic level is 2000 J. **(2)**

13. A doctor has advised Sameer to reduce sugar intake in his diet and do regular exercise after checking his blood test reports. Which disease do you think Sameer is suffering from? Name the hormone responsible for this disease and the organ producing the hormone. **(3)**

14. In pea plant, round seed is dominant over the wrinkled. If a cross is carried out between these two plants, give answer to the following questions. **(3)**

i. Mention the genes for the traits of parents.

ii. State the trait of F1 hybrids.

iii. Write the ratio of F2 progeny obtained from this cross. What is the name of the cross?

15. During a biology class, the teacher showed a diagram of the human heart and explained the double circulation system. Students learned that oxygenated and deoxygenated blood are kept separate, and the heart pumps blood through pulmonary and systemic circuits. The left ventricle has thicker walls than the right because it pumps blood to the entire body.
Answer the following questions based on the case: **(4)**

Attempt either subpart A or B

(A) Why is it important to keep oxygenated and deoxygenated blood separate in mammals?

OR

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(B) Compare the functions of pulmonary artery and pulmonary vein.

(C) What is meant by double circulation in humans?

(D) Which chamber of the heart receives oxygenated blood from the lungs?

16. Attempt either option A or B

(A)Differentiate between the following: (5*1=5)

(a) Pollen tube and style

(b) fission in *Amoeba* and *Plasmodium*

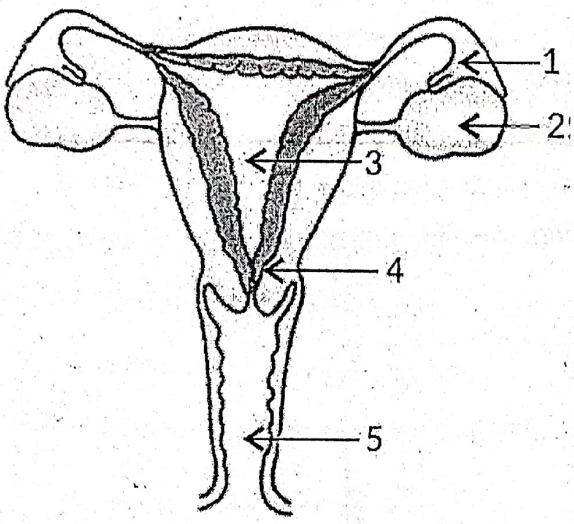
(c) Fragmentation and regeneration

(d) Bud of *Hydra* and bud of *Bryophyllum*

(e) Vegetative propagation and spore formation

OR

(B) (i) Identify the given diagram. Name the parts 1 to 5 (3)



(ii) What are contraceptive devices? Give two reasons for adopting contraceptive devices in humans. (2)

SECTION B

1. Which of the following statements is true for an amphoteric oxide?

a) It reacts with acid as well as base to form salt and hydrogen gas.

b) It reacts only with base and does not form water.

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c) It reacts only with acid and does not form water.

d) It reacts with both acid as well as base to form salt and water (1)

2. $CH_3 - CH_2 - OH \xrightarrow{Alkaline\ KMnO_4+Heat} CH_3 - COOH$

In the above given reaction, alkaline $KMnO_4$ acts as (1)

a) A reducing agent

b) oxidising agent

c) catalyst

d) dehydrating agent

3. In the soap micelles

a) the ionic end of soap is on the surface of the cluster while the carbon chain is in the interior of the cluster. (1)

b) both ionic end and carbon chain are on the exterior of the cluster.

c) ionic end of soap is in the interior of the cluster and the carbon chain is out of the cluster.

d) both ionic end and carbon chain are in the interior of the cluster.

4 .**Assertion (A): Iso - butane is the isomer of C_4H_{10} .**

Reason (R): Iso - butane has four C and ten - H atom.

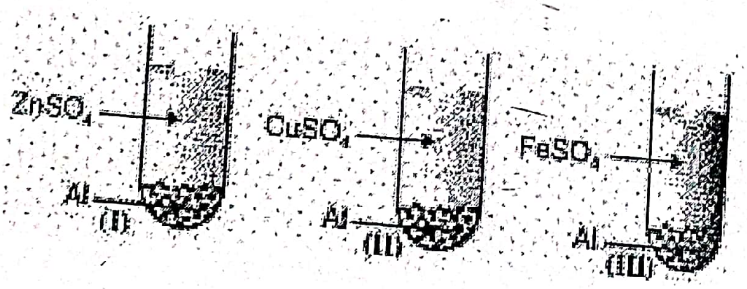
a) Both A and R are true and R is the correct explanation of A. (1)

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true

5.



(1)

Obseravation	I	II	III
Solution after reaction	Colourless	Colourless	Colourless
Metal Deposited	Zn	Cu	Fe

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Which of the following is correct conclusion?

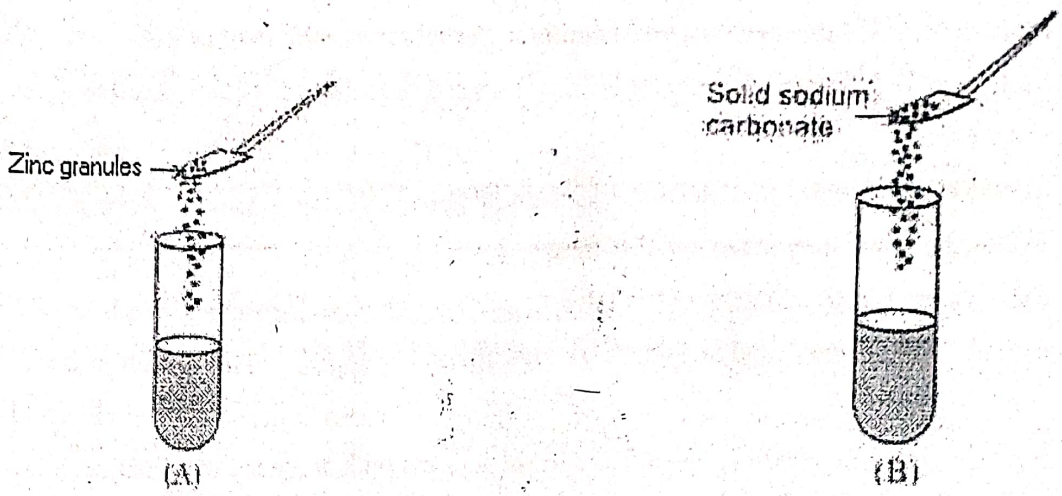
a) Al is more reactive than Zn and Cu but less reactive than Fe

b) Al is more reactive than Cu but less reactive than Zn and Fe

c) Al is more reactive than Cu and Fe but less reactive than Zn

d) Al is more reactive than Zn, Cu, Fe

6.A student took two test tubes containing 2 ml of dilute hydrochloric acid and added zinc granules to test tube (A) and solid sodium carbonate to test tube (B) as shown below: (1)



The correct observation would be

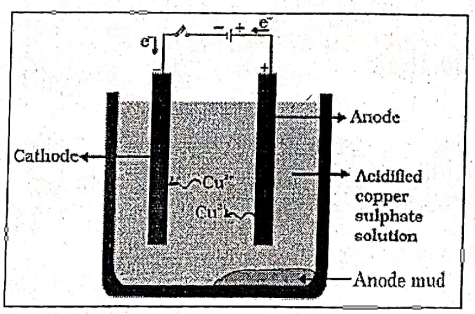
a) Rapid reaction in both the test tubes

b) Rapid reaction in (A) but a slow reaction in (B)

c) No reaction in any of the test tubes.

d) Slow reaction in (A) and rapid reaction in (B)

7.The following diagram shows the electrolytic refining of copper: (1)



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Which of the following statements is **incorrect** description of the process?

a) The impure metal from the anode dissolves into the electrolyte.

b) Insoluble impurities settle down at the bottom of the anode.

c) On passing the current through the electrolyte, the pure metal from the anode dissolves into the electrolyte.

d) The pure metal from the electrolyte is deposited on the cathode.

8. Which of the given statement is correct or wrong: (1)

Statement A: Ethane decolorizes bromine water whereas ethyne does not.

Statement B: Mixture of water and alcohol is used in radiators of vehicles in cold countries.

a) Both - Statement A and Statement B - are false.

b) Statement A is true; Statement B is false.

c) Statement B is true;StatementA is false.

d) Both - Statement A and Statement B - are true.

9. An element A burns with golden flame in air. It reacts with another element B, atomic number 17 to give a product C. An aqueous solution of product C on electrolysis gives a compound D and liberates hydrogen. Identify A, B, C and D. Also write down the equations for the reactions involved. (2)

10. When solutions of silver nitrate and sodium chloride are mixed, white precipitate forms. The ionic equation for the reaction is

$Ag^+(ag) + Cl^- \rightarrow AgCl(s)$ (3)

1.

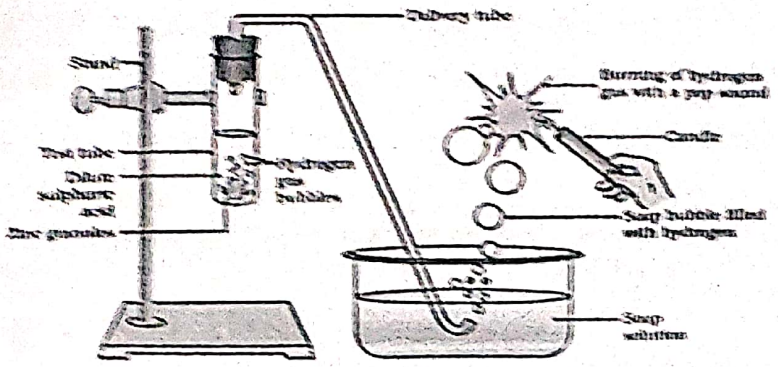
   a. What is the name of the white precipitate?

   b. Is it a soluble or insoluble compound?

2. Is the precipitation of silver chloride a redox reaction?

11. In the following schematic diagram for the preparation of hydrogen gas as shown in fig, what would happen if following changes are made? (3)

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Instead of dilute sulphuric acid, dilute hydrochloric acid is taken.

OR

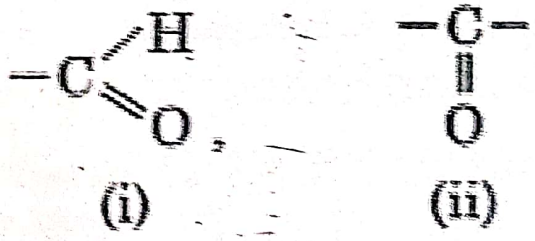
When zinc metal is treated with a dilute solution of a strong acid, a gas is evolved, which is utilised in the hydrogenation of oil. Name the gas evolved. Write the chemical equation of the reaction and also write a test to detect the gas formed.

12.Read the following text carefully and answer the questions that follow:

More than three million carbon compounds have been discovered in the field of chemistry. The diversity of these compounds is due to the capacity of carbon atoms for bonding with one another as well as with other atoms. Most of the carbon compounds are poor conductors of electricity and have low melting and boiling points.

1. Write the molecular formula of first two members of homologous series having functional group - Br. (1)

2. Given below are the formulae of some functional groups: (1)



(i) (ii)

Write the name of these functional groups.

3. What would be observed on adding a 5% alkaline potassium permanganate drop by drop to some warm ethanol taken in a test tube? State the role of $KMnO_4$ in the reaction and write the chemical equation for the reaction involved. (2)

OR

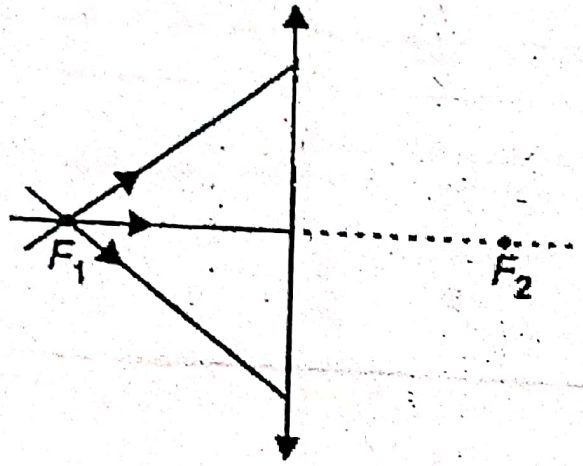
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Write the name of the compound formed when ethanol is heated at 443 K temperature with excess of conc. H_2SO_4. What is the role of conc. H_2SO_4 in the reaction? Write the chemical equation for the reaction involved. (2)

13.Give the steps involved in the extraction of metals of low and medium reactivity from their respective sulphide ores. (5)

SECTION C (PHYSICS)

1.In the given diagram, three rays of light pass through focal point F_1 to the converging lens, placed along vertical line. Which of the following is correct regarding this? (1)



a) The speed of light rays increases when they enter the converging lens.

b) The speed of light rays decreases when they leave the converging lens.

c) The speed of light rays remains same while entering or leaving the converging lens.

d) The speed of light rays decreases when they enter the converging lens.

2.The clear sky appears blue because (1)

a) Light of all other colours is scattered more than the violet and blue colour lights by the atmosphere.

b) Blue light gets absorbed in the atmosphere.

c) Ultraviolet radiations are absorbed in the atmosphere.

d) Violet and blue lights get scattered more than lights of all other colours by the atmosphere.

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3.Assertion (A): The image formed by a concave mirror is certainly real if the object is virtual.

Reason (R): The image formed by a concave mirror is certainly virtual if the object is real.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true. (1)

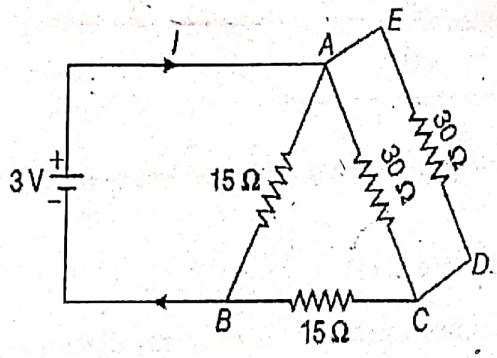
4. Define power of a lens. The focal length of a lens is - 10 cm. Write the nature of the lens and find its power. If an object is placed at a distance of 20 cm from the optical center of this lens, according to the New Cartesian Sign Convention, what will be the sign of magnification in this case? (2)

5.A lamp consumes 50 W and is lighted 2 h daily in month of April. How many units of electric energy is consumed?

Or

Arrange 1Ω , 10 Ω and 100 Ω such that the equivalent resistance is greater than 10 Ω but less than 11 . (2)

6. I. Find the value of current in the circuit given as below:



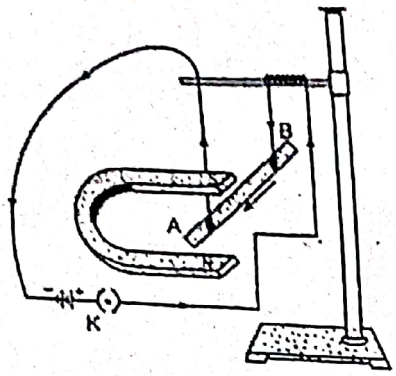
II. You have four resistors of 8Ω each. Show how would you connect these resistors to have effective resistance of 8? (3)

7.I. State two main causes of a person developing near - sightedness. With the help of a ray diagram, suggest how he can be helped to overcome his disability?

II. The far point of myopic person is 100 cm in front of the eye. Calculate the focal length and power of a lens required to enable him to see distant objects clearly. (3)

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8.In activity shown, how do you think the displacement of rod AB will be affected (3)



1. if the current in rod AB is increased
2. astronger horse shoe magnet is used

9.Read the following text carefully and answer the questions that follow: (4)

Study the data given below showing the focal length of three concave mirrors A, B and C and the respective distances of objects placed in front of the mirrors:

Case	Mirror	Focal Length (cm)	Object Distance (cm)
1	A	20	45
2	B	15	30
3	C	30	20

1. In which one of the above cases the mirror will form a diminished image of the object? Justify your answer. (1)

2. List two properties of the image formed in case 2. (1)

3.What is the nature and size of the image formed by mirror C? Draw ray diagram to justify your answer. (2)

OR

An object is placed at a distance of 18 cm from the pole of a concave mirror of focal length 12 cm. Find the position of the image formed in this case. (2)

10.Although electric kettle and electric toaster were used simultaneously in the kitchen to prepare breakfast for the family, yet the two devices could work efficiently due to **fuse** used in the electric circuit. (5)

1. What is a fuse? Write the material used in fuse wires. How is a fuse connected in an electric circuit?

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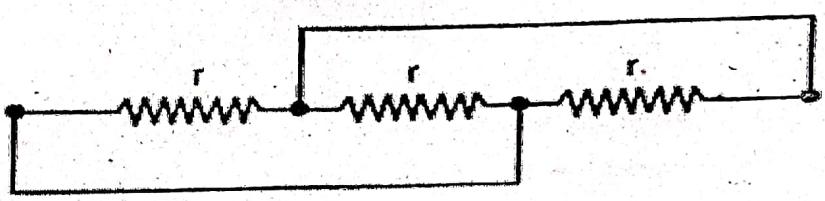
2. State the ratings of fuse used in electric circuits.

3. What is the function of a fuse? How does it perform its function?

4. A device uses 1 kW electric power when operated at 220 V. Calculate the rating of the fuse to be used.

Or

\ Three equal resistors each equal to r and connected as shown in Fig. Calculate the equivalent resistance.



2. State Joules law of heating. How does it depend on the magnitude of current.

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